

Part 4 – Decision Table Testing

Task 4.1 – Decision table for apply_discount()

Condition/Action	R1	R2	R3	R4	R5	R6	R7	R8
discount_type = percentage	Y	Y	N	N	N	Y	N	*
discount_type = flat	N	N	Y	Y	N	N	Y	*
discount_type invalid	N	N	N	N	Y	N	N	*
percentage value 0.01..100	Y	Y	*	*	*	N	*	*
flat value 0.01..invoice_total	*	*	Y	Y	*	*	N	*
status Draft	Y	N	Y	N	*	*	*	N
status Submitted	N	Y	N	Y	*	*	*	N
status Paid/PartiallyPaid/Voided	N	N	N	N	*	*	*	Y
Action: apply discount	Y	Y	Y	Y	N	N	N	N
Action: ValidationError	N	N	N	N	Y	Y	Y	N
Action: InvalidStateError	N	N	N	N	N	N	N	Y

Notes : * = do not care. R1-R4 are valid applications. R5 rejects invalid discount type. R6 rejects invalid percentage values. R7 rejects invalid flat values. R8 rejects non-discountable invoice statuses.